The University of Azad Jammu and Kashmir

Department of Software Engineering

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Roll no	2024-SE-04
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Subject	Software Construction and Development
Course Code	SE-3101
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Lab	01
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Department of Software Engineering

THE UNIVERSITY OF AZAD JAMMU & KASHMIR, MUZAFFARABAD

Department of Software Engineering

Software Construction & Development (Lab) — SE-3101

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Lab Assignment: Lab 1 — Use Case Diagrams

Instructor: Engr. Dr Asma Javaid

Course Code: SE-3101

Semester: Fall 2026

Question 1: Online Ticket Reservation System

1. Problem Statement

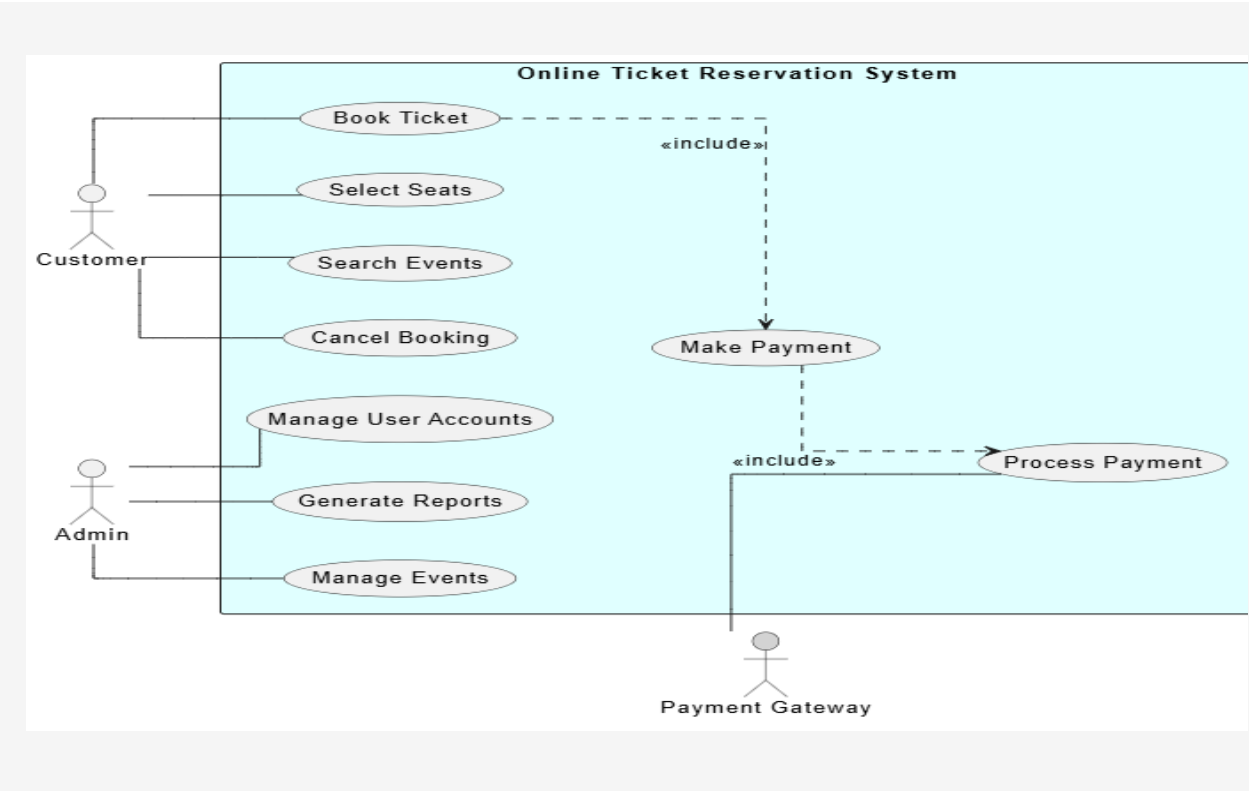
Develop a Use Case Diagram that serves as a basis for understanding system requirements for an Online Ticket Reservation System, identifying all primary actors, administrative functions, and system interactions.

2. Actors & Use Cases Identification

Element Type	Name	Description / Purpose
Actor	Customer	Primary end-user who searches events, selects seats, makes payments, and cancels bookings.
Actor	Admin	Administrative actor who manages event listings, generates statistics reports, and maintains user accounts.
Actor (External)	Payment Gateway	External third-party payment system authorizing and processing card transactions.
Use Case	Search Events	Allows users to browse and filter available events, concerts, and shows.
Use Case	Select Seats	Enables choosing specific seating arrangements and ticket tiers.
Use Case	Book Ticket	Initiates the ticket reservation workflow.
Use Case	Make Payment	Handles payment processing for booked tickets.

Use Case	Process Payment	Executes transaction authorization with the Payment Gateway.
Use Case	Cancel Booking	Allows customers to request reservation cancellations and refunds.
Use Case	Manage Events	Allows Admin to create, update, or remove event listings.
Use Case	Generate Reports	Provides Admin with system sales summaries and attendance metrics.
Use Case	Manage User Accounts	Allows Admin to manage customer profiles and system access.

3. Use Case Diagram



Question 2: Credit Card Processing System

1. Problem Statement

The Credit Card Processing System is used to purchase items from any shop/mall, maintain credit card balance limits, process transactions via a credit card machine, and issue printed balance statements or receipts.

2. Actors & Use Cases Identification

Element Type	Name	Description / Purpose
Actor	Customer	Shopper selecting items, swiping credit card, and signing receipts.
Actor	Vendor	Store staff generating bills and issuing printed balance statements/receipts.
Use Case	Select Item	Customer chooses items/goods to purchase.
Use Case	Create Bill	Vendor generates the invoice for the selected items.
Use Case	Swipe Card & Request Payment	Customer hands over the card to initiate the transaction request.
Use Case	Process Transaction	Card machine verifies credit limits and processes the payment.
Use Case	Issue Balance Statement / Receipt	Vendor hands the printed receipt showing updated balance to the customer.
Use Case	Sign and Return Receipt	Customer signs the paper receipt copy and returns it to the vendor.

3. Use Case Diagram

